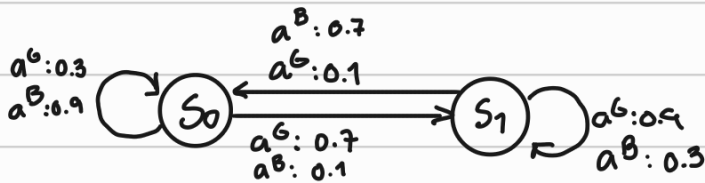


1.

$$P_{aG} = \begin{bmatrix} P_{aG}(S_0|S_0) & P_{aG}(S_1|S_0) \\ P_{aG}(S_0|S_1) & P_{aG}(S_1|S_1) \end{bmatrix}$$



$$Q^\pi(x, a) = C(x, a) + \gamma \sum_{y \in X} P_a(y|x) J^\pi(y)$$

$$\begin{aligned} Q^\pi(S_1, a^G) &= C(S_1, a^G) + 0.9 (P_{aG}(S_0|S_1) V(S_0) + P_{aG}(S_1|S_1) V(S_1)) \\ &= 10 + 0.9 \times (0.1(100) + 0.9(20)) = 35.2 \end{aligned}$$

$$\text{or } J_1 = C a + \gamma P_a J_0$$

$$C(S_0) = \begin{bmatrix} 10 \times 0.3 + 10 \times 0.7 \\ 10 \times 0.1 + 10 \times 0.9 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 10 \\ 10 \end{bmatrix} + 0.9 \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix} \begin{bmatrix} 100 \\ 20 \end{bmatrix} = 35.2$$

$$Q^\pi(S_1, a^B) = 0 + 0.9 (0.7 \times (100) + 0.3 \times (20)) = 68.4$$

2.

$$V^\pi = C^\pi + \gamma P^\pi V^\pi$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix} + 0.9 \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$(\Rightarrow) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix} + \begin{bmatrix} 0.9(0.3x + 0.7y) \\ 0.9(0.1x + 0.9y) \end{bmatrix} (\Rightarrow)$$

$$(\Rightarrow) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 + 0.9(0.3x + 0.7y) \\ 10 + 0.9(0.1x + 0.9y) \end{bmatrix} (\Rightarrow)$$

$$(\Rightarrow) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 + 0.27x + 0.63y \\ 10 + 0.09x + 0.81y \end{bmatrix}$$

$$\begin{cases} x = 10 + 0.27x + 0.63y \\ y = 10 + 0.09x + 0.81y \end{cases} (\Rightarrow)$$

$$(\Rightarrow) \begin{cases} 0.73x = 10 + 0.63y \\ 0.19y = 10 + 0.09x \end{cases} (\Rightarrow)$$

$$(\Rightarrow) \begin{cases} x = \frac{10 + 0.63y}{0.73} \\ 0.19y = 10 + 0.09 \frac{10 + 0.63y}{0.73} \end{cases} (\Rightarrow)$$

$$(\Rightarrow) \begin{cases} x = \text{"} \\ 0.1387y = 7.3 + 0.09(10 + 0.63y) \end{cases}$$

$$\Rightarrow \begin{cases} x = \\ 0.082 \end{cases} \quad y = 8.2$$

$$\Rightarrow \begin{cases} x = \frac{10 + 0.63 \times 100}{0.73} \\ y = 100 \end{cases} \quad \Leftrightarrow$$

$$\Rightarrow \begin{cases} x = 100 \\ y = 100 \end{cases}$$

3.1

$$TV_{\text{new}} = \min_a (C(s, a) + \gamma \sum_{s'} P(s' | s, a) (V_{\text{old}} + C))$$

$$= \min_a (C(s, a) + \gamma P_a V_{\text{old}} + \gamma P_a C) =$$

$$= \min_a (C(s, a) + \gamma P_a V_{\text{old}} + \gamma C) \quad \begin{array}{l} \text{Because the} \\ \text{sum of} \\ \text{the lines} \end{array}$$

$$= \min_a (C(s, a) + \gamma P_a V_{\text{old}}) + \gamma C$$

of P_a is
always 1
so $P_a \times C = C$

$$= TV_{\text{old}} + \gamma C$$

3.2

$$TV_{\text{new}} = TV_{\text{old}} + \gamma C \quad \Leftrightarrow$$

$$\Rightarrow TV_{\text{new}} - TV_{\text{old}} = \gamma C$$

Como $\gamma \in [0, 1[$ a cada iteração

multiplying by γ decrease the difference in between TV_{new} and TV_{old}

$$\left\{ \begin{array}{l} TV_{new} = TV_{old} + \gamma C \\ V_{new} = V_{old} + C \\ TV_{new} = V_{new} \\ TV_{old} = V_{old} \end{array} \right. \quad \begin{array}{l} \Rightarrow C = \gamma C \\ \downarrow \\ \text{At every iteration} \end{array}$$

So Value Iteration is guaranteed to converge